

An abstract averaging method for quasilinear equations

Jean Mawhin¹ and Jorge Novoa²

¹Institut de Recherche en Mathématique et Physique, Université Catholique de Louvain, Louvain-la-Neuve, Belgium

²Department of Mathematical Engineering, Universidad de Chile, Santiago, Chile
`jean.mawhin@uclouvain.be` `jnovoa@dim.uchile.cl`

Abstract

The aim of the paper is to extend the abstract averaging method to nonlinear equations of the form

$$Mx = \varepsilon N(x, \varepsilon),$$

in Banach spaces, allowing the principal operator M to be nonlinear. In particular, M need not admit a linear inverse on a complement of its kernel. The approach is based on a nonlinear complementary correction, which reduces the original equation to an averaged compatibility equation coupled with a nonlinear fixed-point problem. We prove that every nondegenerate zero of the corresponding averaged equation generates, for sufficiently small nonzero values of the parameter, a nearby solution under a compactness assumption. Local uniqueness is obtained under a suitable small Lipschitz condition, while differentiability yields a unique local C^1 branch. We also determine the leading asymptotic profile of the solutions. Applications are given to a vector p -Laplacian Duffing system and to an infinite-dimensional problem in a Hilbert space.

Keywords: Averaging method; quasilinear equations; periodic solutions; ϕ -Laplacian; Duffing systems.

1 Introduction

The averaging method for the periodic problem associated with first-order systems

$$x' = \varepsilon f(t, x, \varepsilon), \quad x(0) = x(T), \tag{1.1}$$

relates the existence of periodic solutions for small $|\varepsilon|$ to the finite-dimensional bifurcation equation

$$\int_0^T f(t, a, 0) dt = 0. \tag{1.2}$$

Under the usual nondegeneracy condition, each solution a^* of (1.2) generates a nearby T -periodic solution of (1.1). This classical principle can be obtained by several small-parameter methods, including the Poincaré operator and the Cesari–Hale approach; see [1, 8] and the references therein.

A particularly direct formulation consists in adjoining to the differential equation the compatibility condition obtained by integration over one period. For $\varepsilon \neq 0$, problem (1.1) is equivalent to

$$x' = \varepsilon f(t, x, \varepsilon), \quad \int_0^T f(t, x(t), \varepsilon) dt = 0.$$

At $\varepsilon = 0$, the second equation selects from the family of constant solutions precisely those satisfying the bifurcation equation. This simple observation leads to an implicit-function formulation of the averaging method; see [9]. A related fixed-point formulation was developed in more general settings in [8], and the same idea was extended in [2] to abstract semilinear equations

$$Lx = \varepsilon N(x, \varepsilon),$$

with L a Fredholm operator of index zero. In that case, the equation is reduced by inverting L on a complement of $\ker L$ and solving the remaining finite-dimensional compatibility equation.

The situation becomes substantially different for genuinely nonlinear principal operators. Periodic equations of the form

$$(\phi(u'))' = \varepsilon f(t, u, u', \varepsilon) \quad (1.3)$$

include the classical p -Laplacian and p -Laplacian-like operators, relativistic acceleration and prescribed mean curvature operators. Their periodic theory has been extensively developed by topological and variational methods; see, for example, [4–6, 10, 11].

In our preceding paper [7], we studied the small-parameter problem (1.3) by a direct implicit-function argument. The method applies when ϕ^{-1} is of class C^1 near the origin and

$$D\phi^{-1}(0)$$

is a continuous isomorphism. It therefore covers, among others, the classical second-order and relativistic cases, and yields a unique local periodic branch near each nondegenerate solution of the corresponding averaged equation.

The linearized invertibility required in that approach, however, excludes some of the most natural quasilinear operators. Consider, for instance,

$$\Phi_p(\xi) = |\xi|^{p-2}\xi, \quad \Phi_p^{-1} = \Phi_{p'}, \quad p' = \frac{p}{p-1}.$$

For $p > 2$, the inverse $\Phi_{p'}$ is not differentiable at the origin, while for $1 < p < 2$ its derivative at the origin is degenerate. Thus, except for $p = 2$, the complementary equation cannot be recovered from an invertible linearization of the principal operator at a constant state. The obstruction is therefore not the bifurcation equation itself, but the absence of the linear complementary inverse on which the usual Lyapunov–Schmidt and implicit-function reductions rely.

The aim of the present paper is to extend the abstract averaging framework of [2] to equations with a nonlinear principal operator, thereby covering genuinely quasilinear problems for which the usual Fredholm reduction is no longer available. At the same time, this removes the linearized invertibility restriction required in our preceding paper [7]. More precisely, we replace the linear complementary inversion by a genuinely nonlinear correction and consider abstract equations

$$Mx = \varepsilon N(x, \varepsilon), \quad 0 < |\varepsilon| \ll 1, \quad (1.4)$$

with decompositions

$$X = E \oplus X_2, \quad Z = Z_1 \oplus Z_2, \quad E = \ker M,$$

and introduce a nonlinear *complementary correction* R . Instead of inverting the linearization of M on X_2 , we require R to solve directly the complementary part of the nonlinear equation. The problem is then reduced exactly to

$$QN(a + v, \varepsilon) = 0, \quad v = R(a + v, \varepsilon), \quad a \in E, \quad v \in X_2.$$

The first equation plays the role of the bifurcation equation, whereas the second retains the nonlinear structure of the principal operator. While nonlinear complementary inverses and quasilinear continuation schemes were already developed in [3, 4], the present reduction is local around a non-degenerate zero of the averaged map and also provides local uniqueness regimes and asymptotic profiles.

This reduction yields three averaging principles. Complete continuity of R gives existence through Leray–Schauder degree; a sufficiently small Lipschitz constant gives local existence and uniqueness by the contraction principle; and, when R is differentiable, the reduced system can again be treated by the implicit function theorem. The construction also determines the leading asymptotic profile whenever the complementary correction has a prescribed small-parameter scaling. In particular, a homogeneous nonlinear inverse of degree γ produces the natural displacement scale $|\varepsilon|^\gamma$, instead of the classical $O(\varepsilon)$ scale.

The final section illustrates two different regimes. A vector p -Laplacian Duffing system with $p > 2$ is treated through compactness, giving multiple periodic solutions and the nonlinear scale $|\varepsilon|^{1/(p-1)}$. We then consider a nonsmooth problem with values in an infinite-dimensional Hilbert space for which the small Lipschitz condition yields local existence and uniqueness.

2 A nonlinear averaging principle

Let X, Z be real Banach spaces and let $M : D(M) \subset X \rightarrow Z$ be continuous. Assume

$$E := \ker M, \quad X = E \oplus X_2, \quad Z = Z_1 \oplus Z_2, \quad \text{Im } M \subset Z_2, \quad (2.1)$$

where all spaces in the decompositions are closed. Let $P : X \rightarrow E$ and $Q : Z \rightarrow Z_1$ be the corresponding projections. Let $U \subset X$ be open and let $N : \overline{U} \times [-\varepsilon_1, \varepsilon_1] \rightarrow Z$ be continuous.

Definition 2.1 (Complementary correction). A continuous map $R : \overline{U} \times [-\varepsilon_1, \varepsilon_1] \rightarrow X_2$ with $R(x, 0) = 0$ is called a *complementary correction* if, for every (x, ε) in its domain, $\varepsilon(I - Q)N(x, \varepsilon) \in \text{Im } M$ and $Px + R(x, \varepsilon) \in D(M)$, and

$$M(Px + R(x, \varepsilon)) = \varepsilon(I - Q)N(x, \varepsilon). \quad (2.2)$$

In addition, every solution x of (1.4) is required to satisfy $R(x, \varepsilon) = (I - P)x$.

Define the averaged map $F : E \cap U \rightarrow Z_1$ by $F(a) = QN(a, 0)$. Fix $a_0 \in E \cap U$ and assume that QN is C^1 near $(a_0, 0)$ and

$$F(a_0) = 0, \quad F'(a_0) = D_x(QN)(a_0, 0)|_E : E \rightarrow Z_1 \quad \text{is an isomorphism.} \quad (2.3)$$

Lemma 2.2 (Nonlinear reduction). Assume that QN is C^1 near $(a_0, 0)$ and that (2.3) holds. Let R be a complementary correction and write $x = a + v$, with $a \in E$ and $v \in X_2$. For $\varepsilon \neq 0$,

$$Mx = \varepsilon N(x, \varepsilon) \iff \begin{cases} QN(a + v, \varepsilon) = 0, \\ v = R(a + v, \varepsilon). \end{cases} \quad (2.4)$$

Moreover, near $(a_0, 0)$ there is a C^1 map $A(v, \varepsilon)$, with $A(0, 0) = a_0$, such that $QN(a + v, \varepsilon) = 0$ if and only if $a = A(v, \varepsilon)$. Hence local solutions are exactly the fixed points of

$$T_\varepsilon(v) := R(A(v, \varepsilon) + v, \varepsilon). \quad (2.5)$$

Set $B_0 := F'(a_0)^{-1}D_x(QN)(a_0, 0)|_{X_2} \in \mathcal{L}(X_2, E)$. Then $D_v A(0, 0) = -B_0$, and for every $\eta > 0$, after shrinking the neighborhood,

$$\|A(v_1, \varepsilon) - A(v_2, \varepsilon) + v_1 - v_2\|_X \leq (1 + \|B_0\| + \eta) \|v_1 - v_2\|_X. \quad (2.6)$$

Proof. If $Mx = \varepsilon N(x, \varepsilon)$, then $Mx \in \text{Im } M \subset Z_2 = \ker Q$, hence $0 = Q(Mx) = \varepsilon QN(x, \varepsilon)$. Thus $QN(x, \varepsilon) = 0$ for $\varepsilon \neq 0$, while Definition 2.1 gives $v = (I - P)x = R(x, \varepsilon)$. Conversely, the two reduced equations and (2.2) give $M(a + v) = \varepsilon(I - Q)N(a + v, \varepsilon) = \varepsilon N(a + v, \varepsilon)$, which proves (2.4).

Set $\Phi(a, v, \varepsilon) = QN(a + v, \varepsilon)$. Since $D_a \Phi(a_0, 0, 0) = F'(a_0)$ is an isomorphism, the implicit function theorem gives the map A and therefore (2.5). Differentiating $\Phi(A(v, \varepsilon), v, \varepsilon) = 0$ at $(0, 0)$ gives $D_v A(0, 0) = -B_0$. The C^1 regularity of A then yields (2.6). $\square$

We shall use, separately, the following three alternative local hypotheses on R near $(a_0, 0)$.

(C) **(Compactness)** R is completely continuous in a neighborhood of $(a_0, 0)$.

(L) **(Small Lipschitz condition)** There exist a neighborhood V of a_0 and $\ell : [0, \varepsilon_1] \rightarrow [0, \infty)$ such that for $x, y \in V$,

$$\|R(x, \varepsilon) - R(y, \varepsilon)\|_X \leq \ell(|\varepsilon|) \|x - y\|_X, \quad \text{and} \quad \limsup_{r \rightarrow 0^+} \ell(r) < \frac{1}{1 + \|B_0\|}.$$

(D) **(Differentiability)** The map $(x, \varepsilon) \mapsto R(x, \varepsilon)$ is of class C^1 in a neighborhood of $(a_0, 0)$ in $X \times \mathbb{R}$.

Theorem 2.3 (Nonlinear averaging principle). *Assume that QN is C^1 near $(a_0, 0)$, that (2.3) holds, and that R is a complementary correction. Then, for every sufficiently small $0 < |\varepsilon|$, the following hold.*

- (i) Under (C), (1.4) has a solution $x_\varepsilon \rightarrow a_0$ in X .
- (ii) Under (L), (1.4) has a unique solution in a fixed sufficiently small neighborhood of a_0 , and $x_\varepsilon \rightarrow a_0$ in X .
- (iii) Under (D), the nearby solutions for $\varepsilon \neq 0$ lie on a unique local C^1 branch $\varepsilon \mapsto x_\varepsilon$ extending to $x_0 = a_0$.

Proof. By Lemma 2.2, local solutions are exactly the fixed points of T_ε in X_2 .

For (i), choose $r > 0$ so that A is defined on $\overline{B}_r^{X_2}(0) \times [-\varepsilon_*, \varepsilon_*]$. Since R is completely continuous, T_ε is compact. After decreasing ε_* if necessary,

$$v \neq \theta T_\varepsilon(v), \quad \|v\|_X = r, \quad 0 \leq \theta \leq 1. \quad (2.7)$$

Indeed, otherwise there would exist $\varepsilon_k \rightarrow 0$, $\|v_k\|_X = r$ and $\theta_k \in [0, 1]$ with $v_k = \theta_k T_{\varepsilon_k}(v_k)$. Complete continuity gives, along a subsequence, $v_k \rightarrow v$ and $\theta_k \rightarrow \theta$; passing to the limit and using $R(x, 0) = 0$ yields $v = 0$, contradicting $\|v\|_X = r$. Hence

$$\deg_{LS}[I - T_\varepsilon, B_r, 0] = \deg_{LS}[I, B_r, 0] = 1,$$

so a fixed point v_ε exists. The same compactness argument shows that any sequence of such fixed points with $\varepsilon \rightarrow 0$ is relatively compact and has only the cluster point 0. Thus $v_\varepsilon \rightarrow 0$ and $x_\varepsilon = A(v_\varepsilon, \varepsilon) + v_\varepsilon \rightarrow a_0$.

For (ii), (2.6) and (L) allow us to choose $q \in (0, 1)$ such that, for small nonzero ε ,

$$\|T_\varepsilon(v_1) - T_\varepsilon(v_2)\|_X \leq q \|v_1 - v_2\|_X.$$

Moreover, $T_\varepsilon(0) = R(A(0, \varepsilon), \varepsilon) \rightarrow R(a_0, 0) = 0$, so a fixed sufficiently small closed ball is invariant under T_ε . Banach's theorem yields a unique fixed point, with

$$\|v_\varepsilon\|_X \leq \frac{\|T_\varepsilon(0)\|_X}{1 - q} \rightarrow 0.$$

The exact reduction gives existence, local uniqueness and $x_\varepsilon \rightarrow a_0$ for (1.4).

For (iii), set

$$\mathcal{G}(a, v, \varepsilon) := (QN(a + v, \varepsilon), v - R(a + v, \varepsilon)).$$

Since $R(x, 0) = 0$, one has $D_x R(a_0, 0) = 0$, and therefore

$$D_{(a,v)} \mathcal{G}(a_0, 0, 0) = \begin{pmatrix} F'(a_0) & D_x(QN)(a_0, 0)|_{X_2} \\ 0 & I \end{pmatrix}.$$

This operator is an isomorphism because $F'(a_0)$ is. The implicit function theorem yields the unique local C^1 branch described in (iii). $\square$

Distinct nondegenerate zeros of F yield distinct local responses by choosing disjoint neighborhoods. A convenient sufficient condition for (L) is $\ell(r) \rightarrow 0$, which occurs, for example, when $R(x, \varepsilon) = \mathcal{R}(\varepsilon(I - Q)N(x, \varepsilon))$ and the relevant maps are locally Lipschitz.

Corollary 2.4 (Asymptotic profile). *Assume that QN is C^1 near $(a_0, 0)$, that (2.3) holds, and that R is a complementary correction. Let $x_\varepsilon \rightarrow a_0$ be a local solution family for $\varepsilon \rightarrow 0^+$, and suppose that, locally uniformly,*

$$\varepsilon^{-\gamma} R(x, \varepsilon) \rightarrow R_0(x), \quad 0 < \gamma \leq 1. \quad (2.8)$$

Put $w_0 = R_0(a_0)$ and $D = D_x(QN)(a_0, 0)$. Then

$$\frac{x_\varepsilon - a_0}{\varepsilon^\gamma} \rightarrow \begin{cases} w_0 - F'(a_0)^{-1} D w_0, & 0 < \gamma < 1, \\ w_0 - F'(a_0)^{-1} (D w_0 + \partial_\varepsilon(QN)(a_0, 0)), & \gamma = 1. \end{cases} \quad (2.9)$$

Proof. From (2.5) and (2.8), $v_\varepsilon/\varepsilon^\gamma \rightarrow w_0$. Moreover,

$$D_v A(0, 0) = -F'(a_0)^{-1} D|_{X_2}, \quad \partial_\varepsilon A(0, 0) = -F'(a_0)^{-1} \partial_\varepsilon(QN)(a_0, 0).$$

The first-order expansion of $A(v_\varepsilon, \varepsilon)$ gives (2.9); for $\gamma < 1$, $\varepsilon = o(\varepsilon^\gamma)$. $\square$

In particular, a positively homogeneous complementary inverse of degree γ changes the classical $O(\varepsilon)$ scale into $O(\varepsilon^\gamma)$.

3 Applications

3.1 A p -Laplacian Duffing system

Let $p > 2$, $\Phi_p(\xi) = |\xi|^{p-2}\xi$ on $\mathbb{R}^n$, and $A = (a_{jk}) \in \mathbb{R}^{n \times n}$. Consider

$$\begin{cases} (\Phi_p(u'))'_j + \varepsilon \left[\sum_{k=1}^n a_{jk} u'_k + b_j u_j^3 + c_j u_j^2 + d_j u_j \right] = \varepsilon e_j(t), & 1 \leq j \leq n, \\ u(0) = u(T), & u'(0) = u'(T), \end{cases} \quad (3.1)$$

where $e \in C_T(\mathbb{R}^n)$, $b_j, c_j, d_j \in \mathbb{R}$, and $c_j = 0$ whenever $b_j = 0$. Put $\bar{e} = \frac{1}{T} \int_0^T e(t) dt$ and $g_j(s) = b_j s^3 + c_j s^2 + d_j s$.

Take $X = C_T^1(\mathbb{R}^n)$, $Z = C_T(\mathbb{R}^n)$, $Mu = (\Phi_p(u'))'$, and $N_j(u) = e_j - \sum_k a_{jk} u'_k - g_j(u_j)$, with the natural domain of M . Then $E = \ker M = \mathbb{R}^n$; let $Pu = \bar{u}$ and $Qh = \bar{h}$. The periodic inverse constructed in [4, Section 2] gives a continuous compact map $\mathcal{K} : Z_2 \rightarrow X_2$ satisfying $M\mathcal{K}h = h$. Hence

$$R(u, \varepsilon) = \mathcal{K}(\varepsilon(I - Q)N(u))$$

is a complementary correction satisfying (C).

For $\alpha \in E$, $F_j(\alpha) = \bar{e}_j - g_j(\alpha_j)$, so the bifurcation equations are

$$b_j \alpha_j^3 + c_j \alpha_j^2 + d_j \alpha_j - \bar{e}_j = 0, \quad 1 \leq j \leq n. \quad (3.2)$$

Since $F'(\alpha) = -\text{diag}(g'_1(\alpha_1), \dots, g'_n(\alpha_n))$, every choice of simple roots in (3.2) gives a nondegenerate zero. Thus, if the j th equation has $m_j \in \{1, 2, 3\}$ simple real roots, problem (3.1) has at least $m_1 \cdots m_n$ distinct T -periodic solutions for all sufficiently small $0 < |\varepsilon|$, each converging to the corresponding constant root.

Moreover,

$$\mathcal{K}(\lambda h) = \text{sgn}(\lambda) |\lambda|^{1/(p-1)} \mathcal{K}h.$$

For a zero α of F , one has $(I - Q)N(\alpha) = e - \bar{e}$ and $D_x(QN)(\alpha)\mathcal{K}(e - \bar{e}) = 0$. Hence Corollary 2.4 gives, as $\varepsilon \rightarrow 0^+$,

$$u_\varepsilon = \alpha + \varepsilon^{1/(p-1)} \mathcal{K}(e - \bar{e}) + o\left(\varepsilon^{1/(p-1)}\right) \quad \text{in } C_T^1(\mathbb{R}^n).$$

3.2 A differential equation in $L^2(0, 1)$

Consider the 2π -periodic problem

$$(\Phi(u'))' + \varepsilon u = \varepsilon e(t), \quad u(0) = u(2\pi), \quad u'(0) = u'(2\pi), \quad (3.3)$$

with $e \in C_{2\pi}(H)$, $H = L^2(0, 1)$, $0 < \lambda < 1$, and $\Phi : H \rightarrow H$ is the Nemytskii operator induced by

$$\rho(s) = s + \lambda |s|,$$

that is,

$$(\Phi v)(x) = \rho(v(x)).$$

Its inverse is induced by

$$\psi(s) = \begin{cases} s/(1 + \lambda), & s \geq 0, \\ s/(1 - \lambda), & s < 0, \end{cases}$$

and therefore, with $\Psi = \Phi^{-1}$,

$$\langle \Psi(\xi) - \Psi(\eta), \xi - \eta \rangle_H \geq \frac{1}{1 + \lambda} \|\xi - \eta\|_H^2, \quad \|\Psi(\xi) - \Psi(\eta)\|_H \leq \frac{1}{1 - \lambda} \|\xi - \eta\|_H. \quad (3.4)$$

Set $X = C_{2\pi}^1(H)$, $Z = C_{2\pi}(H)$, $Mu = (\Phi(u'))'$ and $N(u) = e - u$, with P, Q given by averages. Following the standard periodic inverse construction for ϕ -Laplacian-like operators [4], let $H_h(t) = \int_0^t h(s) ds$ for $h \in Z_2$. The map

$$c \mapsto \int_0^{2\pi} \Psi(H_h(t) + c) dt$$

is continuous, strongly monotone and coercive; hence, by the Browder–Minty theorem [12], it has a unique zero $c(h)$. Set $(\mathcal{K}h)' = \Psi(H_h + c(h))$ and $\overline{\mathcal{K}h} = 0$. Then $M\mathcal{K}h = h$, and (3.4) gives

$$\|\mathcal{K}h_1 - \mathcal{K}h_2\|_{C^1} \leq C_K \|h_1 - h_2\|_\infty.$$

Consequently,

$$R(u, \varepsilon) = \mathcal{K}(\varepsilon(I - Q)N(u))$$

satisfies

$$\|R(u, \varepsilon) - R(v, \varepsilon)\|_{C^1} \leq C|\varepsilon| \|u - v\|_{C^1},$$

so (L) holds. Moreover,

$$F(a) = \bar{e} - a, \quad F'(\bar{e}) = -I_H, \quad B_0 = 0.$$

Hence Theorem 2.3(ii) yields a unique 2π -periodic solution near $\bar{e}$ for all sufficiently small $0 < |\varepsilon|$.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] J.A. Sanders, F. Verhulst, J. Murdock, *Averaging Methods in Nonlinear Dynamical Systems*, 2nd ed., Springer, New York, 2007.
- [2] J.A. Cid, J. Mawhin, M. Zima, An abstract averaging method with applications to differential equations, *J. Differential Equations* 274 (2021) 231–250.
- [3] W. Ge, J. Ren, An extension of Mawhin’s continuation theorem and its application to boundary value problems with a p -Laplacian, *Nonlinear Anal.* 58 (2004) 477–488.

- [4] R. Manásevich, J. Mawhin, Periodic solutions for nonlinear systems with p -Laplacian-like operators, *J. Differential Equations* 145 (1998) 367–393.
- [5] R. Manásevich, J. Mawhin, Boundary value problems for nonlinear perturbations of vector p -Laplacian-like operators, *J. Korean Math. Soc.* 37 (2000) 665–685.
- [6] J. Mawhin, Periodic solutions of systems with p -Laplacian-like operators, in: *Nonlinear Analysis and its Applications to Differential Equations*, Birkhäuser, Boston, 2001, pp. 37–63.
- [7] J. Mawhin, J. Nova, An averaging method for periodic solutions of quasilinear equations in Banach spaces, preprint, 2026.
- [8] N. Rouché, J. Mawhin, *Ordinary Differential Equations: Stability and Periodic Solutions*, Pitman, Boston, 1980.
- [9] J. Mawhin, Functional analysis and boundary value problems, in: J.K. Hale (Ed.), *Studies in Ordinary Differential Equations*, MAA Studies in Mathematics, vol. 14, Mathematical Association of America, Washington, DC, 1977, pp. 128–168.
- [10] J. Mawhin, Resonance problems for some non-autonomous ordinary differential equations, in: *Stability and Bifurcation Theory for Non-Autonomous Differential Equations*, Lecture Notes in Mathematics, vol. 2065, Springer, Heidelberg, 2013, pp. 103–184.
- [11] M. García-Huidobro, R. Manásevich, J. Mawhin, S. Tanaka, Periodic solutions for nonlinear systems of ODEs with generalized variable-exponent operators, *J. Differential Equations* 388 (2024) 34–58.
- [12] E. Zeidler, *Nonlinear Functional Analysis and its Applications, II/B: Nonlinear Monotone Operators*, Springer, New York, 1990.